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Digital Forensic Investigation of Cyberbullying through Deepfake Image and Social Media Abuse

Scenario Background

This project presents a fictional cyberbullying investigation designed to simulate a realistic digital forensic case in a controlled academic environment. In the scenario, a university student named **Bell** becomes the victim of an online defamation campaign after a malicious actor operating under the alias **letsleakkarma** publishes a manipulated (deepfake) image portraying Bell as intoxicated in a public location. The image is subsequently distributed through fake social media accounts and an anonymous online discussion forum with the intention of damaging the victim's reputation.

The investigation recreates techniques commonly observed in real-world cyberbullying incidents, including the use of fabricated online identities, manipulated digital media, and coordinated dissemination across multiple online platforms. All social media accounts, websites, images, and digital evidence used throughout the project were intentionally created for educational purposes, ensuring that no real individuals were targeted or harmed. This controlled scenario provides a realistic environment for demonstrating digital forensic investigation techniques, image forensic analysis, evidence preservation, and open-source intelligence (OSINT) methodologies while maintaining ethical and legal considerations.

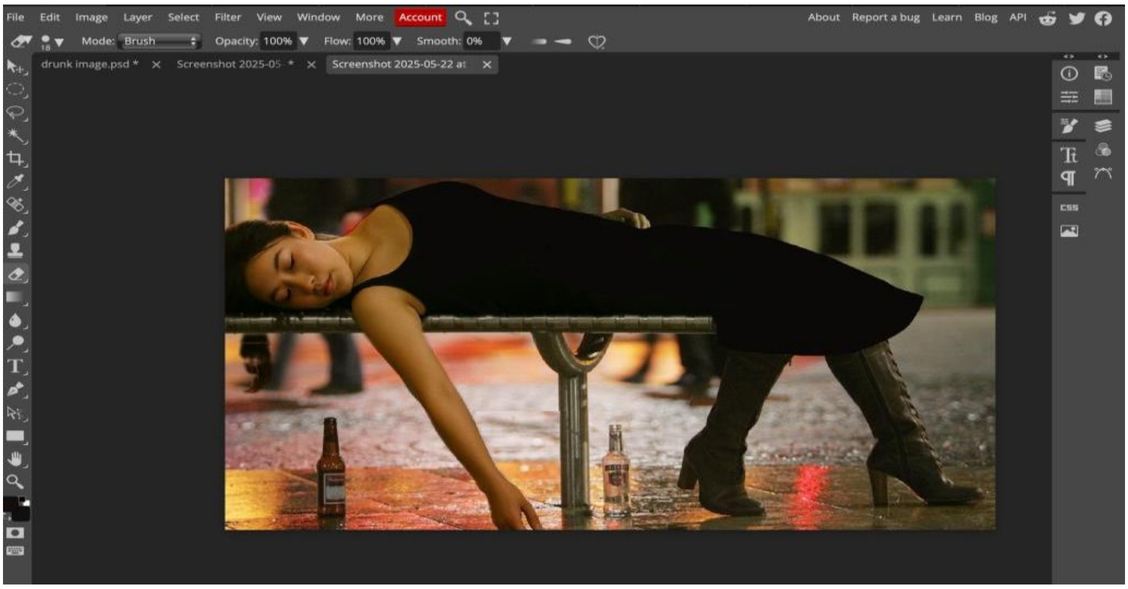
Investigation Objective

The objective of this project was to simulate the investigation of a cyberbullying incident involving manipulated digital media and anonymous online activity. The investigation aimed to demonstrate how digital forensic tools and open-source intelligence (OSINT) techniques can be used to identify indicators of image manipulation, trace an attacker's online footprint, collect and preserve digital evidence, and document investigative findings following accepted forensic principles.

The project also sought to demonstrate proper evidence handling through metadata analysis, reverse image searching, forensic imaging, cryptographic hashing, and structured documentation while maintaining an ethical approach by using only simulated accounts, AI-generated media, and fictional identities.

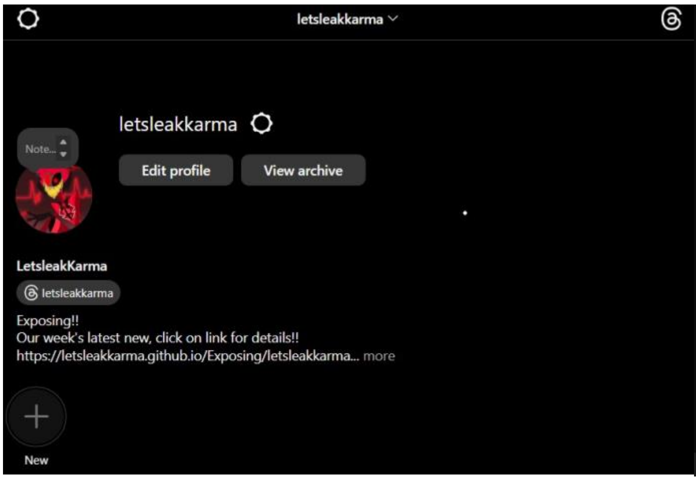
- ## Creating and editing the image

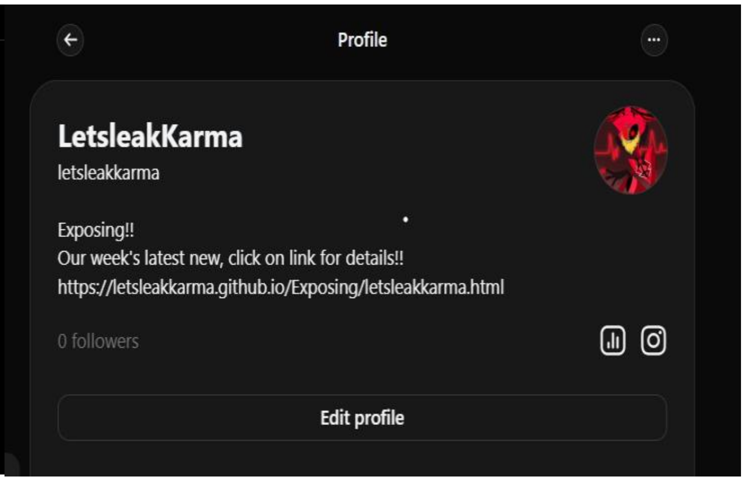
I edited the AI generated image by using Photopea, an image editor. The womans picture was seamlessly integrated into a setting showing an inebriated individual sitting on a park bench in public place for the purpose of replicating common deepfake methods.The software tools employed for this task involved layering effects,scale adjustments,image rotations. Aligning the background appropriately.



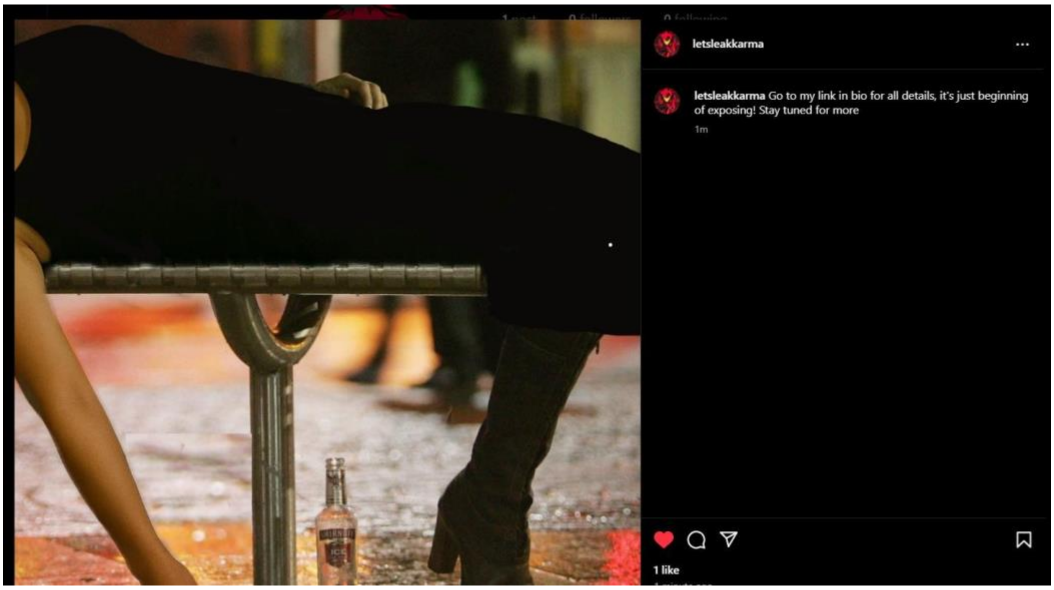
- ## Fake Social Media Accounts and Forums

I created a new Instagram and named it as **letsleakkarma** to act as an attacker. The biography was written to guide individuals to an HTML page hosted on GitHub. I made sure to use the username, for both my Threads account and GitHub profile to keep things in terms of identity.

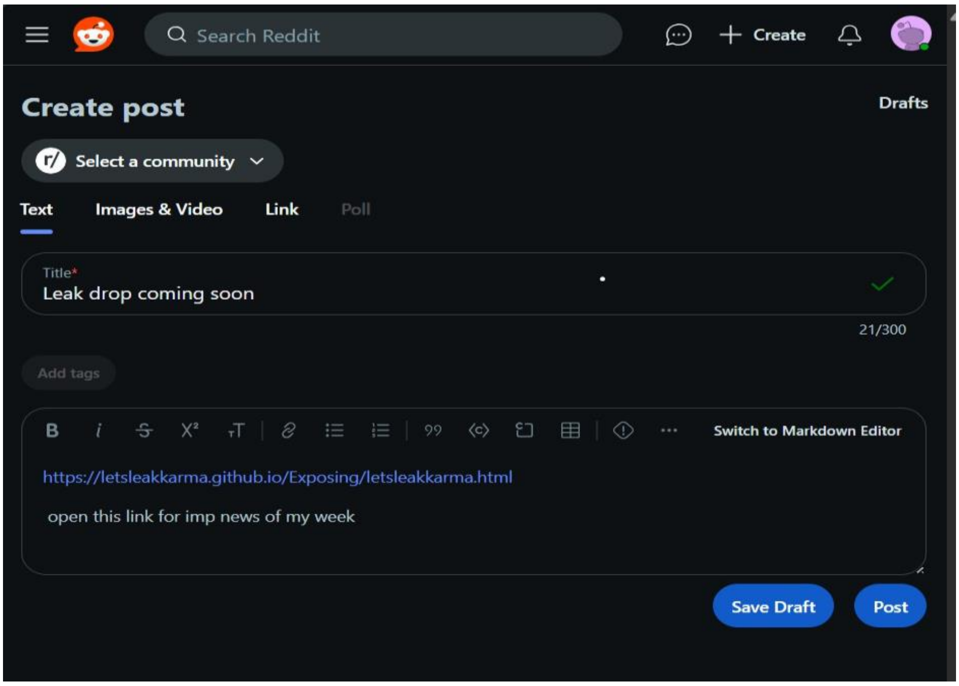




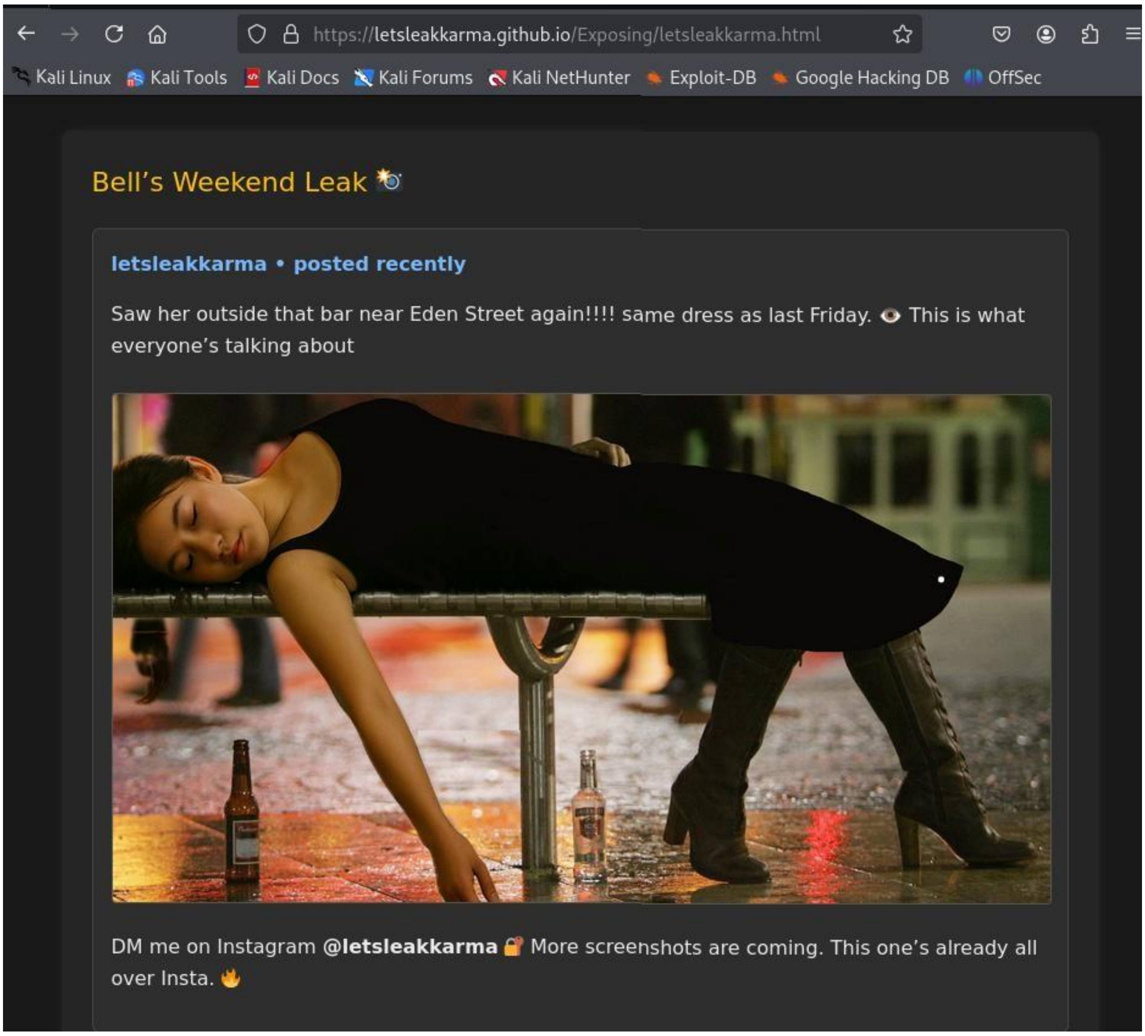
Instagram Post



Reddit Thread



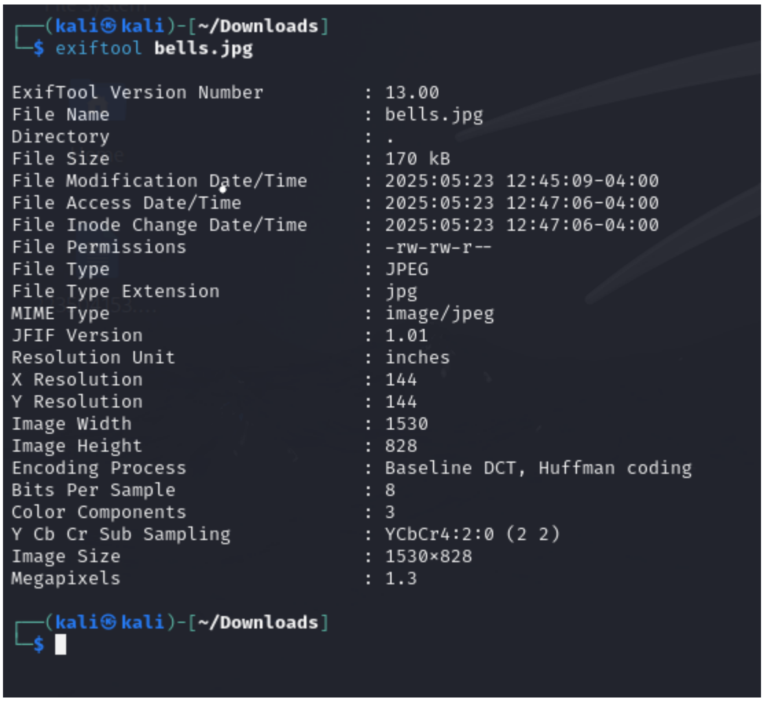
The HTML code was customized with CSS to give it the look and feel of a discussion platform, with usernames and post formatting for a touch. This simulated forum was stored in a GitHub repository linked to the "letsleakkarma". Then shared the link through social media profiles as part of a strategy resembling how an actual attacker might distribute harmful content online.



- **Forensic Analysis of Image**

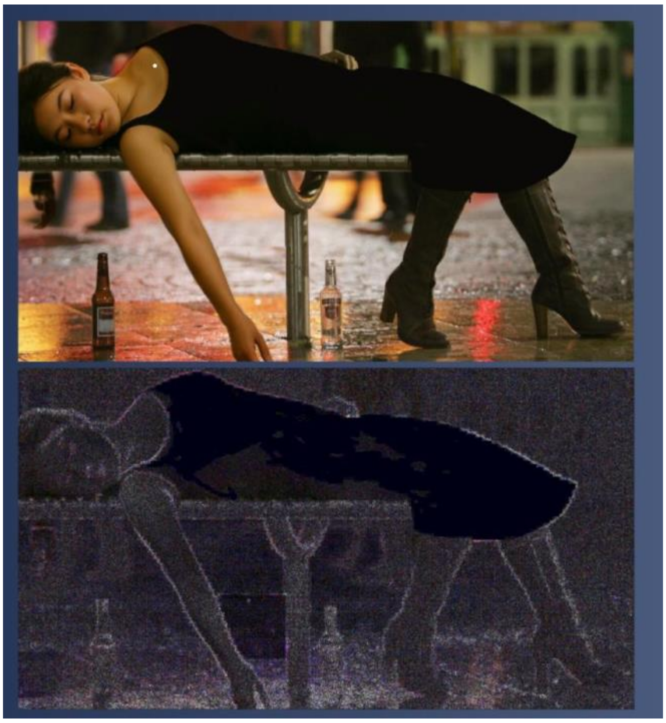
- **ExifTool**

I downloaded the image from forum and started the forensic analysis of the image by using multiple tools. First, I used the Exiftool to see the metadata of the file, examination uncovered characteristics, like file size, quality settings and compression specifics. However, crucial metadata such, as the camera model and creation timestamps appeared to be absent or modified hinting at tampering or re saving of the file using editing tools.



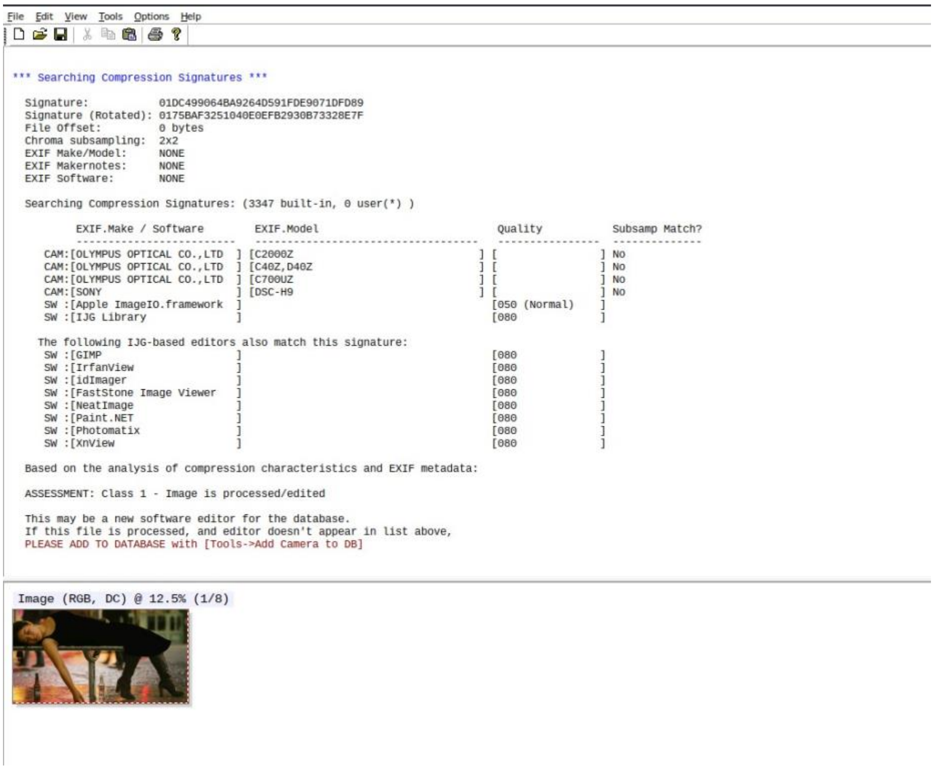
- **FotoForensics ELA (Error Level Analysis)**

These discrepancies strongly suggest that the image has been digitally manipulated, supporting the suspicion of a deepfake scenario.

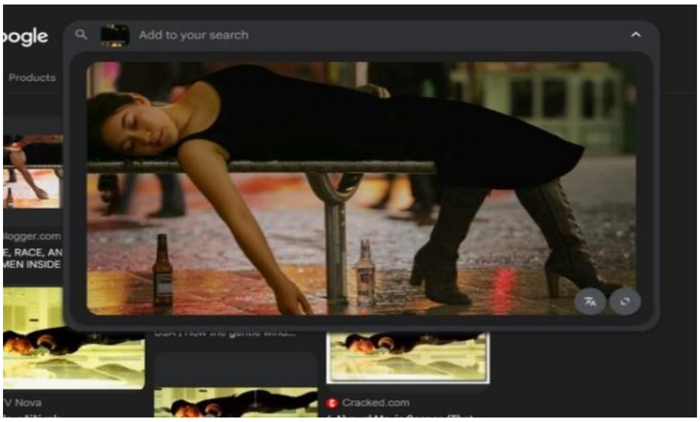


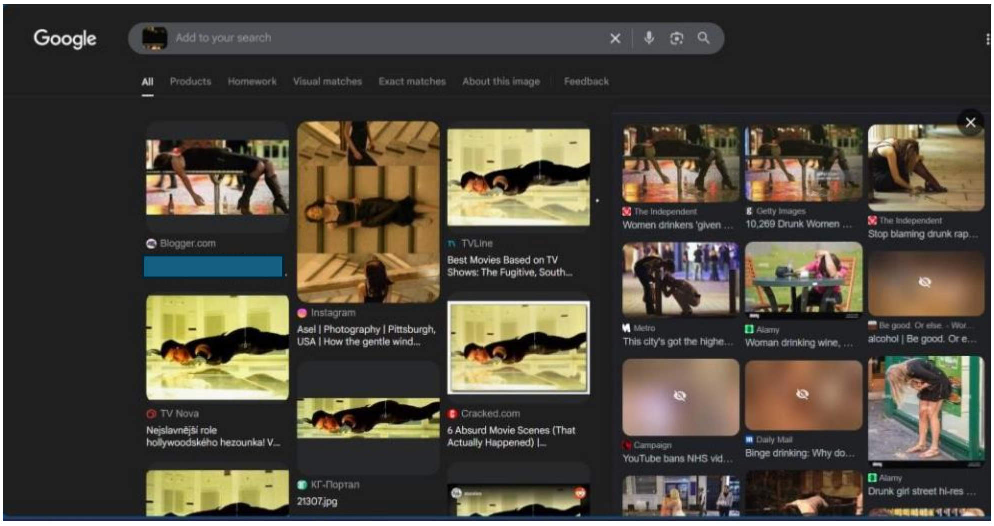
- **JPEGsnoop Examination**

  JPEGsnoop was used to analyse the JPEG compression signatures to determine whether the image was likely captured directly by a digital camera or had been modified using image-editing software. The analysis returned an **"Assessment: Class 1 – Image is processed/edited"**, indicating that the image did not match the compression profile of a known camera and had likely undergone post-processing. Although JPEGsnoop cannot conclusively identify the specific software used to edit an image, it identified compression characteristics consistent with common image-editing applications, including GIMP. These findings support the conclusion that the image was digitally manipulated before being distributed by the attacker.



Reverse Image Search Results and Attribution



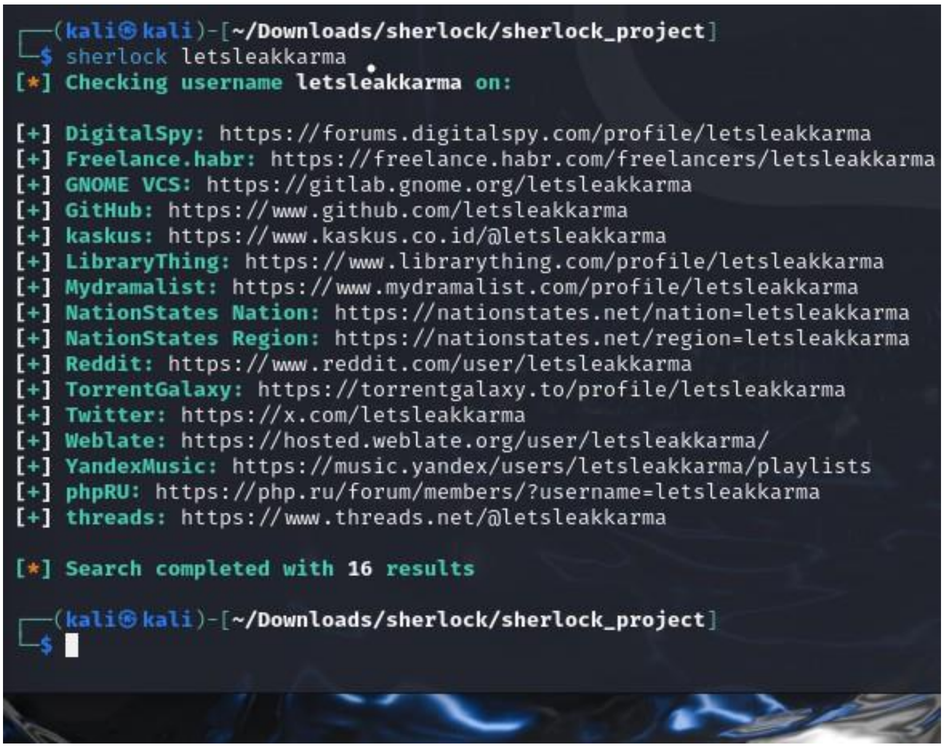


During the examination process I conducted an image search, on Google to confirm the source of the modified image aimed at the victim named Bell. The search resulted in a match on The Independent website, a known news outlet. The picture was initially featured in an article titled " portrayal of drinkers" (dated September 4th 2011) discussing how women are depicted in nightlife situations, by the media. This discovery indicates that the offender probably took the picture from a media source that's available, to the public and modified it to portray the victim highlighting how publicly accessible media can be manipulated in deepfake situations to create deceptive storylines.

- **OSINT Investigation Sherlock**

To uncover the trail left by the offender known as letsleakkarma I turned to Sherlock. A freely available OSINT tool created for tracking down usernames on various social media and internet platforms.

sherlock letsleakkarma



Sherlock was used to search for the username **letsleakkarma** across more than 300 supported websites. The search identified **16 matching accounts**, including profiles on several well-known platforms.

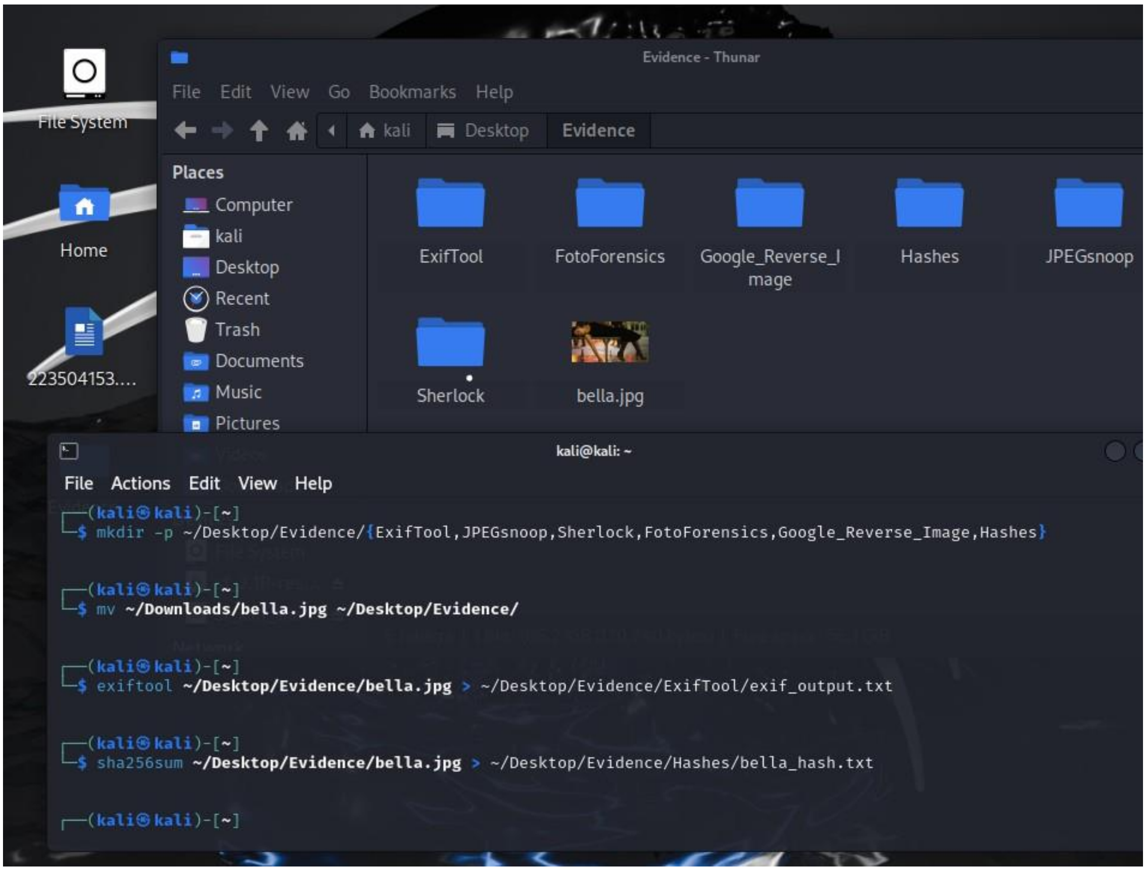
Key findings included:

- The simulated HTML forum created by the attacker was hosted on GitHub and linked to the **letsleakkarma**username.
- Matching usernames were identified on platforms such as Reddit, X (formerly Twitter), and TorrentGalaxy, demonstrating how the same alias could be used across multiple platforms to distribute harmful content.
- Additional matches were found on platforms such as LibraryThing and Kaskus, indicating that the username was also present on unrelated services.

The investigation demonstrated how an attacker could use a consistent online alias (**letsleakkarma**) across multiple platforms to establish a recognisable digital identity and increase the visibility of a coordinated cyberbullying campaign.

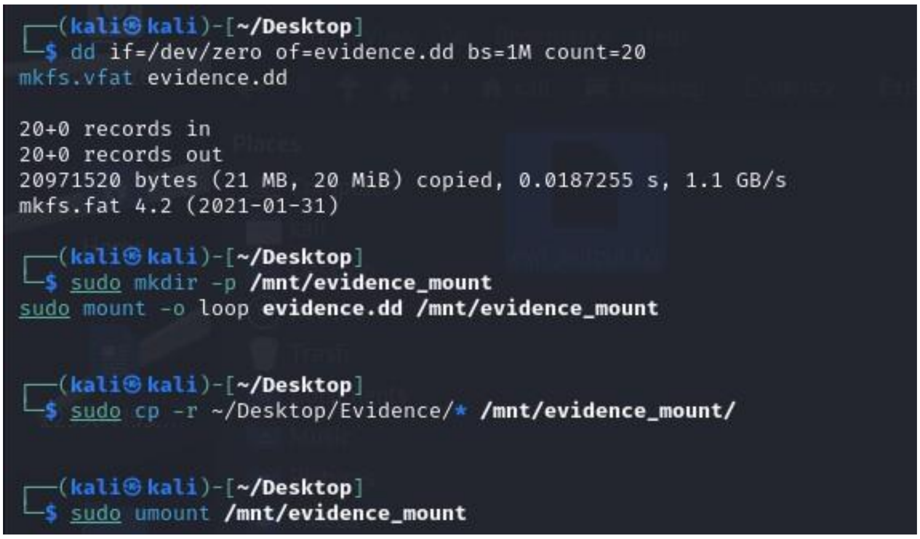
However, Sherlock has several limitations. The tool only identifies the presence of a username on supported websites and does not verify account ownership, account activity, profile authenticity, or whether the accounts belong to the same individual. Consequently, some matches, such as those found on LibraryThing or NationStates, may represent unrelated users, inactive accounts, or placeholder profiles. Therefore, Sherlock results should be treated as investigative leads and corroborated with additional evidence before drawing conclusions.

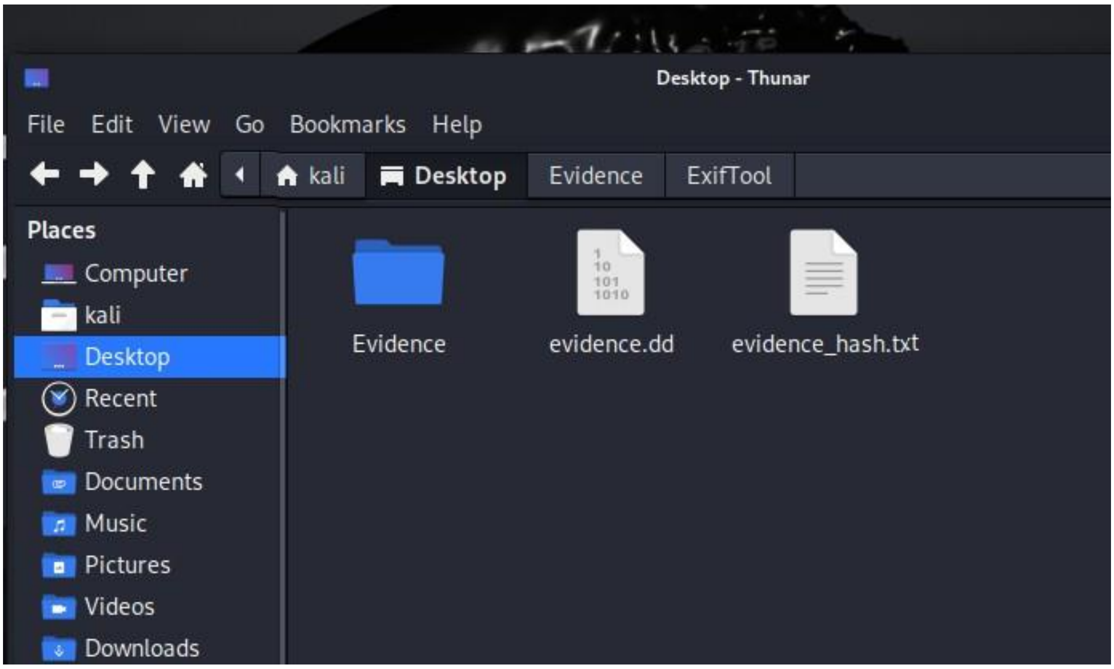
I followed a systematic procedure to maintain digital forensic best practices while protecting evidence integrity through the preservation of manipulated images and metadata reports and OSINT outputs.



The investigation materials included Sherlock outputs together with Exif Tool image metadata and JPEG snoop results and reverse image search screenshots and manipulated images which were stored in the Evidence/ folder. The Evidence/ folder functioned as a single location to store all digital evidence from the simulated cyberbullying case.

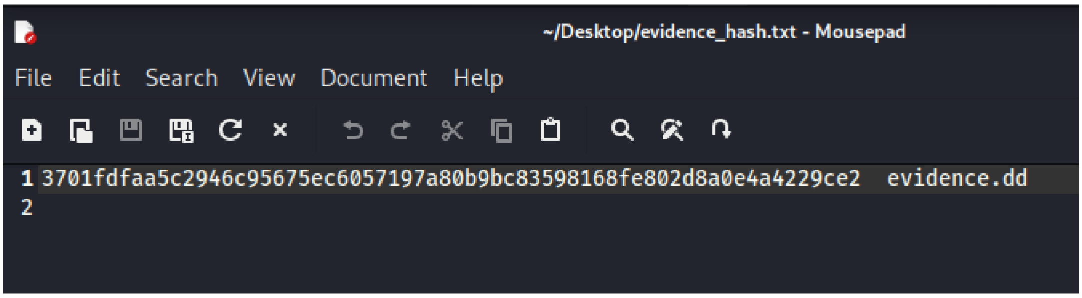
To further maintain the authenticity of data I created the raw disk image **evidence.dd** of my Evidence folder. This approach replicates the process of world imaging to guarantee the preservation of evidence, in a non-modifiable format.





To safeguard the image against tampering and confirm its integrity I created a SHA256 hash of the.dd file.

sha256sum evidence.dd > evidence_hash.txt



- ## Conclusion

  This project successfully recreated a simulated cyberbullying investigation involving a manipulated (deepfake) image and anonymous online harassment within a controlled academic environment. The scenario demonstrated how digital evidence can be collected, analysed, and preserved to investigate online abuse while adhering to ethical and legal considerations. A range of digital forensic and OSINT tools were used throughout the investigation.

  ExifTool, FotoForensics, JPEGsnoop, and Google Reverse Image Search were employed to identify image manipulation, examine metadata, and trace the origin of the manipulated image. Sherlock was used to identify online accounts associated with the alias letsleakkarma, demonstrating how open-source intelligence techniques can assist investigators in identifying an individual's digital footprint across multiple platforms. To maintain the integrity of the evidence, all digital artefacts, including screenshots, metadata, forensic outputs, and recovered files, were securely preserved using forensic imaging and SHA256 hashing. These procedures ensured that the evidence remained authentic and verifiable throughout the investigation. Overall, the project demonstrated the value of combining digital forensic methodologies with OSINT techniques to investigate cyberbullying incidents involving manipulated media. It also highlighted the importance of ethical handling of simulated digital evidence and illustrated how a structured forensic approach can support the identification, preservation, and analysis of evidence in modern cybercrime investigations.

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